36104 · ASSESSMENT 1

Critique and Repair

See what a chart claims · repair it with the Visual Vocabulary · verify a machine's reading of its data

Weight	30% of the subject · individual
Covers	Sessions 1–2: *Seeing Data* and *Choosing Visual Forms*
Artefacts	Three published charts from three different domains — one **primary**, two **supporting**; choices recorded by Monday 10 August
Released	Seeing Data session — Thursday 30 July 2026
Supervised component	25-minute AI-restricted window at the start of the Dashboards-session studio — Thursday 13 August 2026
Due	**Monday 24 August 2026, 11:59 pm** (Sydney time)
Submit	One completed `A1_template.ipynb` · your data files in `data/` · a PNG/screenshot of each original chart

The task

You will take **three published charts from the wild — from three different domains** — and work them through the course pipeline: *see* what each claims, *repair* it with the Visual Vocabulary, and *verify* a machine's interpretation of the data. One artefact is your **primary** (full treatment); the other two are **supporting** (compact treatment). The first two sessions rehearse every part of this: the Seeing Data studio (Critique and Repair sheet, Critique 1) rehearses Parts A and C; the Choosing Visual Forms studio (Redesign ×3, Chart Choice Decision Record) rehearses Part B.

Part A — Seeing (the critiques)

For the **primary artefact**: the full 9-field critique — what does the chart show, and what does it claim? Selection, encoding, framing, omission — keeping the discipline from Seeing Data: *directly visible* vs *interpretation rather than observation*, ending with a repaired caption. **Drafted by hand in the supervised, AI-restricted window on 13 August**, then transcribed into the notebook's `critique_1` cell. For each **supporting artefact**: a compact 4-field critique (main claim, visible vs interpretation, what misleads, repaired caption).

Part B — Choosing (the repairs)

For each of the three artefacts: obtain or faithfully reconstruct its underlying data and produce **one redesign** — the chart that original should have been. Across the three artefacts the redesigns must serve **three distinct Visual Vocabulary categories**. Defend the primary redesign with a full **Chart Choice Decision Record** (the same form as the Choosing Visual Forms studio, including the rejected alternative and the specific reason); defend each supporting redesign with its category and a one-line justification (`WHY_2`, `WHY_3`).

Part C — Verifying (the claim audit, primary artefact)

Generate an interpretation of your **primary** dataset using the **supplied prompt** (verbatim, in the template). Classify **every claim** with the four-way taxonomy from the Seeing Data lab:

Label	Meaning
`supported`	the data directly backs it — attach an evidence probe
`plausible but unverified`	sounds right; this dataset cannot settle it
`unsupported`	asserted with no evidence in this data either way
`contradicted`	the data shows otherwise — attach an evidence probe

Probes are zero-argument functions returning the evidence (a filtered or aggregated DataFrame/Series) — required for every `supported` and `contradicted` claim. Finish with the five-question AI disclosure.

Your artefacts

Choose **three artefacts from three different domains**: at least two from the **released, verified pool**; at most one may be your own found-in-the-wild chart (a real published chart, a named publisher and date, and an obtainable or reasonably reconstructable data source — approval required). Record your three choices, and which is your **primary**, by **Monday 10 August 2026** (Canvas). Two students may study the same source chart, but every submission must be individual and independently produced.

Data rule. If the exact data is not published, reconstruct it faithfully (digitise, or synthesise to match the chart's visible quantities) and say so: set `data_status = "reconstructed"` and write the reconstruction note. This is the atlas book's own method — representative data, never passed off as the original measurement. Undisclosed reconstruction is an academic-integrity issue. Disclosed reconstruction remains eligible for full marks; its fidelity and limitations are judged under H2. The rule applies per artefact (`data_status` in each `META_n`).

Copyright note. Include only the chart image needed for criticism and review, cite its publisher, creator (where known), date and source URL, and do not republish the assessment beyond the subject site. Follow any additional UTS library or copyright guidance supplied with the task.

AI policy

Part A's primary critique is drafted AI-free in the supervised window. An assistant may be used in Part B under the notebook contract: docstrings first, accept or reject completions, and nothing counts until the checks pass. Part C requires an assistant-generated interpretation; students who cannot or do not wish to use an assistant can request a course-supplied generated interpretation. The disclosure block is assessed: report use accurately, including legitimate non-use in Part B. False or misleading disclosure is an academic-integrity issue.

Submission contract (read this twice)

The template notebook defines named variables and functions per artefact (`META_1..3`, `critique_1..3`, `data_1..3`, `redesign_1..3`, `decision_record`, `claims` … `disclosure`). The autograder executes your notebook top-to-bottom on a clean machine and inspects those names. **Run the self-check cell before submitting** — it is the same code the marker runs. A notebook that does not execute end-to-end scores 0 on the automated items until resubmitted (one resubmission, capped at 80%).

Rubric — 100 marks (scaled to 30% of the subject)

Automated (60 marks, scored by `a1_autograder.py`)

ID	Item	Marks	Check
G0	Notebook executes end-to-end	6	all cells run clean, no errors
G1	Three artefacts: metadata valid, domains distinct	6	`META_1..3` fields present; URLs well-formed; `data_status` valid; three **distinct** domains
G2	Data provenance $\times 3$	9	each `data_n` loads; ≥ 6 rows $\times \geq 2$ columns; no empty columns; reconstruction note ≥ 30 words when reconstructed
G3	Redesign mechanics $\times 3$	15	each redesign returns its Axes; three **distinct** Vocabulary categories, all valid; bar-family charts include a zero baseline; every chart titled with ≥ 15 characters
G4	Claim-audit structure (primary artefact)	14	interpretation ≥ 400 chars from the supplied prompt; ≥ 6 claims; every label from the four-way taxonomy; ≥ 1 `supported`, ≥ 1 `plausible but unverified`, ≥ 1 `unsupported` **or** `contradicted`; an executable, non-empty probe for every `supported` / `contradicted` claim
G5	Critiques + Decision Record structure	6	full 9-field primary critique, 5–150 words each; compact critiques 2–3 complete; all 9 Decision Record fields, defence fields ≥ 10 words; `WHY_2` / `WHY_3` ≥ 10 words
G6	Five-question AI disclosure	4	all 5 answers completed, none placeholder
	Total	**60**	

Human-marked (40 marks, anchored 0/1/2 per item)

Markers read the autograder report first, then judge only these five items.

Item (weight)	0	1	2
H1 · Critique insight, primary (×5)	Restates the chart	Correctly identifies the claim and one real omission	Also separates *directly visible* from *interpretation*, names the framing device, and the repaired caption actually repairs
H2 · Reconstruction fidelity, across artefacts (×2.5)	Data unrelated to the original's quantities	Plausible magnitudes and units	Matches the original's visible values, or deviations explicitly disclosed
H3 · Decision Record + supporting defences (×5)	No argument, or category label only	Correct task category with a generic justification	Ties audience → task → channel precision; the rejected alternative is real and the reason specific
H4 · Redesign craft, across the three (×5)	Unreadable or misleading	Honest defaults, readable	Titles state findings; scales honest; labels carry the argument
H5 · Audit judgement (×2.5)	Classifications mostly wrong	Right labels, thin probes	Probes genuinely discriminate — a `contradicted` probe could have exonerated the claim; `plausible but unverified` used precisely, not as a dodge
Total: 40 marks (weights × anchor score; maximum $2 \times (5 + 2.5 + 5 + 5 + 2.5) = 40$)			

Process requirements (not marked, but gates)

- Part A supervised draft handed in during the 13 August window (paper or photo). Missing draft → H1 capped at 1.
- Three artefact choices (and the primary) recorded by Monday 10 August; approval recorded for any bring-your-own.
- One resubmission permitted for G0 failures, capped at 80% overall.

What good looks like

A strong A1 reads like the class notebooks: the critique names the claim in one sentence and the repaired caption says what the original's should have said; the Decision Record argues from the reader's task rather than from taste, and its rejected alternative is one you genuinely considered; the reconstruction note is boringly specific; at least one probe shows a fluent-sounding claim to be `contradicted`; and the disclosure says exactly what the assistant drafted, what you rejected, and how you caught its mistakes.

Questions: raise them in the studio or contact the subject coordinator. AI use in this assessment is governed by the AI policy above; unauthorised use is academic misconduct under UTS rules.